

Final Verification, Code Review & Deployment Readiness Report

Date: 2026-07-29

Project: Hamro Hospital Management System

Branch: arena/019fa6fc-hamrohospital

Verification Result

Status: PASSED after final fixes.

The application was reviewed for role-based access, core navigation, reports, exports, migration readiness, and deployment configuration. Existing UI/design was preserved.

Commands Executed

```
python manage.py check
python manage.py migrate --noinput
python manage.py makemigrations --check --dry-run
python manage.py test workflow reports -v 1
DJANGO_DB_ENGINE=postgres python manage.py check
python manage.py create_demo_accounts
```

Results:

```
System check identified no issues
No migrations pending after final migration cleanup
Ran 9 tests OK
PostgreSQL settings check passed
Demo accounts created/updated successfully
```

Smoke-Tested Pages

Public pages:

```
/
/about/
/departments/
/doctors/
/medical-services/
/extension-services/
```

Admin/authenticated operational pages:

```
/accounts/dashboard/super-admin/
/admin/
/accounts/staff/
/accounts/staff/attendance/
/accounts/staff/attendance/punch/
/accounts/staff/leave/
/accounts/staff/salary/payments/
/accounts/system-readiness/
/manage/departments/
/manage/doctors/
/patients/dashboard/
```

/patients/search/
/patients/list/
/billing/
/billing/search/
/laboratory/
/laboratory/queue/
/radiology/
/radiology/queue/
/pharmacy/
/pharmacy/prescriptions/
/admissions/
/admissions/wards/
/nursing/
/operation-theatre/
/operation-theatre/surgeries/
/blood-bank/
/insurance/
/finance/
/finance/extension-fees/
/reports/
/workflow/orders/
/workflow/payments/
/workflow/revenue/
/referrals/queue/
/medical-records/

Report pages:

/reports/registrations/
/reports/departments/
/reports/doctors/
/reports/counters/
/reports/finance/
/reports/department-revenue/
/reports/pharmacy/
/reports/laboratory/
/reports/radiology/
/reports/admissions/
/reports/insurance/
/reports/nursing/
/reports/operation-theatre/
/reports/blood-bank/
/reports/staff-attendance/
/reports/staff-leave/
/reports/payroll/
/reports/staff-departments/

All smoke-tested pages returned HTTP 200. Department Head and Staff denial checks returned redirects as expected.

Export Verification

PDF and Excel exports were verified for:

Overall revenue
Registration
Cash counter
Finance
Department revenue
Pharmacy
Laboratory
Radiology
Admission
Insurance
Staff attendance
Staff leave
Payroll
Department-wise staff

All tested exports returned HTTP 200 with correct PDF or XLSX content types.

Role Verification

Verified demo role login/dashboard access:

admin -> Super Admin dashboard
registration -> Patient registration dashboard
ehs -> EHS/extension registration dashboard
cashier -> Billing dashboard
doctor -> Doctor dashboard
pharmacy -> Pharmacy dashboard
laboratory -> Laboratory dashboard
radiology -> Radiology dashboard
insurance -> Insurance dashboard
admission -> Admissions dashboard
nursing -> Nursing dashboard
operationtheatre -> Operation Theatre dashboard
bloodbank -> Blood Bank dashboard
accounts -> Finance dashboard
medicalrecords -> Medical Records dashboard
staff -> Staff profile
departmenthead -> Staff profile / department-scoped access

Permission spot checks:

Department Head blocked from finance reports
Department Head blocked from payroll reports
Department Head blocked from salary payment management
General Staff blocked from finance reports

Completed Modules and Workflows

- Role-based staff authentication and dashboards

- Patient registration and barcode-driven lookup
- OPD visit/token workflow
- EHS/Extension Service registration workflow
- Patient portal visit booking/profile/password/timeline
- Doctor consultation workflow
- Doctor to Laboratory workflow
- Doctor to Radiology workflow
- Doctor to Pharmacy/prescription workflow
- Doctor to Admission recommendation/referral workflow
- Doctor to Blood Bank/Nursing/OT referral workflow
- Billing and cash counter workflow
- Pending bill/payment workflow
- Staff discount via staff barcode workflow
- Admission deposit credit workflow
- Finance extension-fee management
- Department service order queue
- Notifications and referral routing
- Patient timeline workflow
- Pharmacy batch/stock ledger workflow
- Laboratory structured results and verification
- Radiology templates and verification
- Admission/ward/bed/discharge checklist workflow
- Nursing notes and vitals workflow
- Operation Theatre surgery workflow
- Blood Bank unit/request/issue workflow
- Insurance claim workflow
- Staff attendance/manual punch/device API/CSV import
- Leave request and approval workflow
- Payroll and salary slip workflow
- Comprehensive reports with PDF/XLSX exports
- Audit log foundation
- Production setup scripts and environment examples
- SQLite local development and PostgreSQL production readiness

Issues Fixed During Final Verification

- Added final clean migrations so ``makemigrations --check --dry-run`` reports no pending model changes.
- Added backward-compatible doctor workspace URLs to prevent old ``/doctors/dashboard/`` style links/bookmarks from returning 404.
- Reconfirmed PDF/Excel report exports after final changes.
- Rebuilt final ZIP packages with the latest source and documentation.

Remaining Items Requiring Real Deployment or External Services

These are not code-completion blockers but require real infrastructure/credentials:

1. Real fingerprint machine SDK/device integration. Current alternatives: manual punch, device punch API, and CSV import.
2. Live eSewa/Khalti/FonePay merchant testing with real gateway credentials.
3. Production Redis server for WebSocket/Channels deployment.
4. Celery worker/beat services for scheduled background jobs.

5. Production PostgreSQL database credentials and server provisioning.

6. Full hospital user acceptance testing with real operational data.

Recommendations

- Use PostgreSQL in production with regular automated backups.
- Keep DEBUG=False and configure ALLOWED_HOSTS/CSRF_TRUSTED_ORIGINS for production domains.
- Store secrets in `.env`, never in Git.
- Use HTTPS behind Nginx/Apache reverse proxy.
- Add Redis + Celery workers for production notifications/background jobs.
- Schedule daily database and media backups.
- Add real payment gateway UAT before accepting live digital payments.
- Add real fingerprint SDK integration only after the device brand/model is finalized.
- Run role-based regression tests before every deployment.

Final Conclusion

The Hamro Hospital Management System is verified as stable for local/demo deployment and prepared for production deployment. The final codebase includes integrated workflows, role-based permissions, barcode-driven patient/staff identification, billing, reporting, attendance, payroll, department modules, PDF/Excel exports, and PostgreSQL-ready configuration while preserving the existing website and dashboard UI.